CATS INSIGHTS

Real Curiosity. Real Writing. Real Kent.

IMPACT REPORT

for the RSC Kent Young Chemist Award 2026

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Section A: Executive Summary

CATS Insights is a student-led academic publication founded by Kyal Sin (Jason) Ye Yint at The Worthgate School, Canterbury, Kent. Established in the 2024–25 academic year, the organisation was created to address a specific gap: students at the school had no structured platform through which to engage with genuine academic research, develop scientific writing competencies, or experience the rigour of the professional editorial process. CATS Insights was built to fill that gap, not as a school club, but as an institutional platform with professional standards, formal governance, and a succession plan designed to outlast its founder.

The organisation now comprises 37 active members, supported by a library of 48 foundational governance documents spanning seven suites, from constitution and bylaws to editorial handbooks, quality assurance protocols, and succession planning. It operates a complete academic publishing infrastructure: citation protocols, quality standards, a five-stage editorial pipeline, and structured mentoring through iterative feedback cycles.

For the purposes of this award, CATS Insights provides evidence across all five RSC categories: teaching resources that have been produced; a history of chemistry research series rooted in Kent's landscape and industry; two STEM school events delivered to the wider student body; a structured mentoring programme with quantified impact data; sustained digital outreach; and a dedicated evidence portfolio website. The following sections detail the evidence, the metrics, and the institutional systems that underpin this work.

Section B: RSC Category Mapping

The table below maps each RSC award category to the specific CATS Insights activity, deliverable, and evidence location. Every activity listed has been completed, documented, and archived.

RSC Award Category Mapping

RSC Category	Category Name	Activity
1	Teaching Resource	Original high quality educational materials and teaching resources, engineered by and for students
2	History of Chemistry	An ongoing series of feature articles exploring the science hidden in Kent's iconic landmarks, industry and heritage.
3	STEM Workshop Events	Writing Chemistry for Non-Chemists and UNESCO Learning Planet STEM Workshop Events
4	Mentoring	Quantified measurable success in academic development of students via mentoring and providing guidance, feedback and support and collecting data on their performance
5	Digital Outreach	Various platforms for digital outreach to share our work and fufill our promise of accessible excellence

Section C: Teaching Resource (Category 1)

Several figures were created, diagrams of processes or molecules that were not readily available on the internet that were required to be created for some other goals.

Furthermore, I have architected several training workshop courses, complete with scripts, lesson slides and recordings for most. These courses cover academic writing, research, design and every skill that goes into working on the CATS Insights publication. Specifically curated to train students from having no experience or skills in academic writing and editing into fully prepared writers and editors and designers for CATS Insights, these mastercourse teaching resources are readily available.

Section D: History of Chemistry (Category 2)

The Chemistry of Kent is a series of three long-form feature articles connecting A-Level and beyond-syllabus chemistry to the history, industry, and landscape of Kent. Each article is written to Level 2 standard (accessible to an educated non-specialist), follows the CATS Insights citation protocol, and has been audited through the full CATS Insights editorial and QA pipeline to verify the publish-ready standard.

Article 1: "Written in White: The Chemistry of Dover's White Cliffs"

This article explores the calcium carbonate geochemistry of the Dover cliffs, covering biogenic sedimentation by coccolithophores during the Cretaceous period, the diagenesis of chalk, thermal decomposition reactions, and acid–base chemistry. It connects the geological heritage of Kent's most iconic landmark to A-Level inorganic chemistry, offering students a tangible local context for abstract syllabus content. The article runs to approximately 2,900 words and includes original figures illustrating the chemical processes described.

Article 2: "Six Scientists, Seven Drugs: The Pfizer Sandwich Story"

This article chronicles the medicinal chemistry legacy of Pfizer's research site at Sandwich in Kent — one of the most productive pharmaceutical laboratories in British history. The article leads with fluconazole rather than the more commonly cited sildenafil: a deliberate editorial decision designed to signal deeper engagement with the pharmacological significance of the site. It covers drug discovery pipelines, structure–activity relationships, and the economic impact of pharmaceutical research on the local community. The article examines the work of six scientists across seven major drug discoveries, contextualising Kent's contribution to global public health.

Article 3: "Brewed in the Garden: The Chemistry of Kentish Hops"

This article examines the organic chemistry of hop cultivation and brewing in the Garden of England. It covers the thermal isomerisation of alpha-acids during the boil (the conversion of humulone to iso-humulone that produces bitterness), Maillard reactions in malt chemistry, and the volatile terpene profiles (myrcene, linalool, geraniol) responsible for hop aroma. The article connects Kent's agricultural heritage to A-Level organic chemistry through the lens of a 400-year brewing tradition. It includes original molecular diagrams and a comprehensive glossary of 75 technical terms.

All three articles were also developed into Instagram carousel design guides for digital distribution. The Pfizer carousel (v4.1) and Hops carousel (v3.0) have been audited and cleared to design-ready standard, translating key chemistry content into an accessible visual format for broader audiences. The series is structurally parallel: each article follows the same section architecture, referencing depth, and audit standard, so the series reads as a coherent body of work rather than three disconnected pieces.

Section E: STEM School Events (Category 3)

Two workshops were prepared, delivered, and documented as part of the CATS Insights academic development programme, extending the organisation's impact beyond its membership to the broader student body at The Worthgate School.

Workshop 1: "Writing Chemistry for Non-Chemists"

A foundational STEM communication masterclass teaching students to distinguish between three audience levels (Level 1: specialist; Level 2: educated non-specialist; Level 3: general public) and to write chemistry at an appropriate level of accessibility. Students were introduced to the Three-Test Framework (Specialist Test, Peer Test, Citation Test), the PEEL paragraph structure, and strategies for controlling jargon without sacrificing technical accuracy. The workshop included practical exercises where attendees rewrote technical passages to different audience levels and self-assessed their work using the framework.

Workshop materials were purpose-built and are publicly downloadable from the CATS Insights website: a Workshop Presentation Plan, the 3-Level Framework Handout, a Topic Menu Handout, a "Fix This Paragraph" Exercise Sheet, Level 2 Writing Submissions tracker, and the Workshop Presentation Slides. Physical exercise sheets completed by attendees are archived as timestamped evidence.

Workshop 2: UNESCO Learning Planet Day Workshop

On the request of The Worthgate School, CATS Insights prepared and delivered a workshop for UNESCO's Learning Planet Day — an internationally registered school event. The session covered academic writing methodology, research skills, how to read and extract information from research papers without plagiarising, and a masterclass spanning multiple academic disciplines. This event demonstrated that the school's own administration recognised CATS Insights as a credible source of academic training for the wider student body.

Section F: Mentoring & Measurable Impact (Category 4)

CATS Insights operates a structured mentoring system through its five-stage editorial pipeline. Every student writer submits work through a formal process: initial draft submission, Lead Editor review using the 3-Level Writing Framework and PEEL checklist, written feedback returned via Microsoft Teams using a standardised Feedback Summary Template, writer revision and resubmission, and final review for editorial readiness. This is a documented, systematic process, not informal peer support, designed to produce measurable improvement in academic writing competency.

23	14	78%	+76%
Members Mentored	Sessions Delivered	Avg. Attendance Rate	Avg. Score Improvement

The impact of this system is tracked through formal assessment instruments. CATS Insights developed and deployed 11 mark scheme instruments across a Writer Course and an Editor & QA Course. All metrics are computationally validated from raw data collected between October 2025 and March 2026.

Assessment Progression: Case Studies

Three case studies were constructed from verified assessment data in the Master Metrics Tracker, representing the range of improvement trajectories observed in the programme:

Student	Improvement	Focus Area	Verified Feedback Extract
Student A	+70% over baseline	Level 2 formatting transition	Demonstrated strong ability to translate expert-level density into an educated non-specialist format while retaining scientific precision. Minor formatting rules still missed.
Student B	+80% over baseline	Jargon simplification	Made an excellent strategic decision to completely translate complex jargon (endothelial dysfunction, atherogenesis) into plain English rather than just defining them.
Student C	+60% over baseline	Citation and authority integrity	Made a massive leap forward by integrating in-text citations linked to a credible bibliography. Tendency toward dense run-on sentences when defining every technical term.

These feedback extracts are sourced directly from the Level 2 Writing Assessments logged in the Master Metrics Tracker. Each score maps to a specific mark scheme instrument administered at a specific point in the member's development. The data was collected systematically as part of the editorial workflow, not retrospectively assembled for this submission.

Beyond formal assessment, editorial mentoring extended to the blog article pipeline. The President (Kyal Sin Ye Yint) personally edited all member-submitted blog articles across multiple iterations, providing line-by-line feedback on technical accuracy, audience calibration, paragraph structure, and referencing.

Several members arrived with no prior experience in academic writing; through sustained, structured feedback, their work reached a standard suitable for publication.

Section G: Digital Outreach (Category 5)

CATS Insights maintains a sustained, multi-platform digital presence designed to extend the reach of its chemistry content beyond the school gates. Digital outreach is treated as an integral part of the mission, not an afterthought.

The Evidence Portfolio Website

The CATS Insights website (catsinsights.netlify.app) was designed, built, and deployed by Kyal Sin (Jason) Ye Yint as a permanent digital platform. It serves as both the public-facing portal for the organisation's work and the evidence hub for this submission. The site includes a homepage with the organisation's mission and RSC category mapping, an about page detailing team structure, governance, and the foundational document suite, a blog section hosting the full Chemistry of Kent articles with other articles from other students, a resources page with all training course materials, a mentoring page with assessment data and case studies, a workshop page with downloadable materials, and a digital outreach page.

The website is not merely a delivery mechanism for evidence, it is evidence. A professional, branded website with structured content, downloadable resources, embedded privacy controls, and chemistry-focused articles constitutes direct proof of digital outreach using current technology. The site's full source code is published on GitHub (https://github.com/jasonyeyint/cats-insights-website) for complete transparency.

Instagram (@catsinsightsofficial)

The Instagram account operates in Professional mode for access to Insights analytics. It publishes academic content including carousel summaries of the Chemistry of Kent series, educational infographics, organisational updates, workshop coverage, and team introductions (all of these are completed, we can't post yet due to consent and privacy rules where we have to collect everyone's and their guardians' consent in writing, but we have already created everything).

Additional Digital Platforms

Platform	Handle / URL	Function
YouTube	@catsinsightsofficial	Workshop recordings, course walkthroughs, long-form educational video content (same privacy issues so currently empty)
GitHub	https://github.com/jasonyeyint/cats-insights-website	Full source code of the evidence portfolio website, open, transparent, version-controlled
Email	catsinsightscanterbury@gmail.com	Official contact for collaboration, press, and correspondence
Website	https://catsinsights.netlify.app	The evidence portfolio itself, built entirely by Kyal Sin (Jason) Ye Yint for CATS Insights

Section H: Academic Development Courses

CATS Insights does not expect members to produce publication-standard work without training. Before any member enters the editorial pipeline, they complete structured academic development courses designed and authored by Kyal Sin (Jason) Ye Yint. Three courses have been fully delivered, with three more in development.

Delivered Courses

Course	Scope	Content
Writer Course	10 lessons, 9 scripts, 4 assessments	Research methodology, source evaluation, citation protocols, the 3-Level Writing Framework, PEEL paragraph structure, audience calibration, and scientific writing best practices
Editor & QA Course	8 lessons, 8 scripts, 5 assessments	Constructive feedback methodology, colour-coded scoring system, Turnitin and GPTZero interpretation, citation verification, and the complete QA audit procedure
Designer Course	6 lessons, 6 scripts, 1 assessment	Transforming data and dense text into compelling, accurate infographics and diagrams that enhance science communication. Visual identity standards and brand compliance

Upcoming Courses

Course	Scope	Content
Research Tutorial Videos	10 videos, 102 minutes	JSTOR, Boolean search, Google Scholar, MyBib, reading papers, source evaluation, citation verification
CATS Skills	3 courses, 7 lessons	Achievement logging (STAR method), DofE Award logging (SMART goals), responsible AI use policy
CATS Honours	6 lessons, 6 hours	University application strategy, evidence quantification, narrative construction, interview preparation (UK and US tracks)

Course delivery was deliberately distributed across the senior leadership team. Members were trained in the course content by Kyal Sin (Jason) Ye Yint and then asked to deliver sessions themselves, a strategy to build institutional teaching capacity and ensure the knowledge does not reside with a single individual. All lesson recordings are archived (cannot be openly released due to aforementioned privacy issues).

Section I: Governance, Sustainability & Institutional Legacy

What distinguishes CATS Insights from a typical school club is the infrastructure built to ensure it outlasts its founder. The organisation is governed by 48 foundational documents organised across 7 suites, each subject to four quality tests: the Clarity Test, the Completeness Test, the Standalone Test, and the Consistency Test.

Suite	Focus	Contents
Suite 1	Governance & Constitution	8 documents: Constitution, delegation framework, role descriptions with KPIs, succession playbook, code of conduct, annual planning, decision-making protocol, organisational chart
Suite 2	Academic Development	9 documents: Writer's guide, citation protocol, research handbook, topic menu (30+ pre-approved topics), 3-Level Framework guide, research planning worksheet, draft template, academic integrity declaration, FAQ
Suite 3	Editorial & QA	8 documents: Editorial handbook, 20-point editorial checklist, QA gatekeeper protocol, feedback templates, AI policy, common errors guide, fact-checking protocol, editor training curriculum
Suite 4	Design & Visual Identity	8 documents: Master style guide (colours, typography, logo rules), publication design manual (12-column grid), social media guide, poster protocol, data visualisation standards, template library index, image guidelines, designer curriculum
Suite 5	Operations & Administration	9 documents: Operations manual, meeting protocol, master database guide, new member onboarding, new captain onboarding, comms templates, digital workspace guide, end-of-year handover, attendance policy
Suite 6	Production Pipeline	6 documents: Production workflow diagram (6-stage lifecycle), T-Minus deadline calendar, RFD handoff protocol, pre-publication checklist, edition planning guide, post-publication review
Suite 7	Special Projects & External	8 documents: Proposal template, budget template, risk assessment, partnership agreement, workshop planning pack, certificate system (Bronze/Silver/Gold), application guide, CGS expansion pack

The Succession Test

> *"If every current member left tomorrow, could a new team publish Edition 1 within six months using only our documents?"*

This is the benchmark against which the entire Foundational Suite is evaluated. Every leadership role has a designated deputy. A structured presidential transition protocol ensures knowledge transfer. The three-tier organisational structure distributes authority and accountability so that no single point of failure exists in the operational chain.

The Quality Assurance Pipeline

Every article published by CATS Insights passes through a four-stage audit: (1) editorial review against the 3-Level Writing Framework and PEEL structure; (2) Turnitin plagiarism screening at a threshold below 15%; (3) GPTZero AI detection screening with manual authorship verification via Google Docs version

history and other forms of evidence; and (4) citation verification against primary sources. Articles have been returned for revision at every stage. The standard exists to be enforced, and it is enforced.

Section J: Student Voices

The following testimonials are sourced directly from CATS Insights members and reflect their experience of the organisation's training, editorial process, and culture. Of course more can be found on the website.

> *"CATS Insights is an organisation that supports young creatives, providing resources to produce high-quality articles and providing the opportunity to express their creative selves."* — Elain, Year 12 — Lead Designer

> *"The writing course provided by CATS Insights was very beneficial for me. It helped me become more knowledgeable and improve my writing skills, and I believe this skill will benefit me in the future."* — Elif, Year 12 – Writer

> *"CATS Insights has taught me the depth of academic work and has developed the curiosity in my A-Levels outside of my lessons, enabling me to write about my subjects outside of the specification!"* — Philip, Year 12 – Lead Writer

Section K: Closing Statement

CATS Insights was built on a straightforward conviction: given structure, direction, and purpose, a student's intellect will always exceed what anyone expects of it. The organisation exists to provide students with the platform, the mentoring, and the institutional infrastructure to produce work of genuine academic excellence, and to do so in a way that connects directly to the science of their local community in Kent.

This submission does not rest on a single activity or a single category. It presents a systemic approach to the promotion of education among young people: accessible and high quality teaching resources; a history of chemistry series rooted in Kent's landscape, industry, and heritage; workshops that teach scientific communication to students who have never experienced it; a mentoring pipeline that produces measurable, data-backed improvement in academic writing and research; and a sustained digital presence that extends the organisation's reach well beyond the school gates. Each element reinforces the others.

This is not a personal project submitted for a credential. It is an organisation that promotes academic excellence, develops talent, and democratises access to the experience of rigorous academic publishing. It was built to last. And it will.

Kyal Sin (Jason) Ye Yint
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Year 12 | The Worthgate School, Canterbury, Kent
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